

A Sharp Quarter-Rate Threshold for a First-Derivative Interpolation Dimension Count

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Abstract

At rate $1/4$, the first-derivative interpolation dimension comparison has exact infimum agreement threshold $(3 + \sqrt{133})/31$. This holds for arbitrary monomial jet supports closed under translation in the zeroth jet, using exact saturated local rank, and extends to the stated translation-stable nonmonomial jet spaces. We also account for the exact finite-length weights and coefficient cutoff: in the full-coefficient monomial model, even multiplicity growing with block length cannot lower the asymptotic threshold. Finite-length corrections permit only a constant number of agreement coordinates of improvement. For historical comparison, optimizing monomial supports or translation-stable nonmonomial jet spaces cannot improve the cubic list-budget and fifth-power MCA-budget dependence in the September 10 draft’s specified reconstruction formulas. Separately, for the current ePrint’s full-coefficient monomial order ideals and either graded row or column-only test, we prove necessary quadratic list-budget and fourth-power regular-MCA-budget dependence. These finite-length cost statements use an explicit corrected margin, uniformly in multiplicity; the row converse requires downward closure in both jet coordinates, whereas the column converse only requires it in Y_0 . An explicit high-rate support optimization gives the sharp continuum threshold above half rate, improving a known sufficient curve at every rate above $8 - 3\sqrt{6}$. Two exact finite certificates instantiate this improvement. These are statements about a specified dimension-counting method, not intrinsic lower bounds on Reed–Solomon lists or proximity gaps.

Version distinction. The current 145-page ePrint 2026/2056 [2] proves first-order list and MCA bounds $O_\rho(n/\eta_1^2)$ and $O_\rho(n^2/\eta_1^4)$, respectively. Section 7’s cubic/fifth-power budget converse concerns only the explicitly stated older reconstruction formulas. It does not establish tightness of these current bounds. Section 8 separately proves the current powers necessary for coordinatewise monomial order ideals with full derivative-weighted coefficient prefixes, using either the graded row or column-only test; the latter also covers supports downward only in Y_0 . The result uses this note’s stronger local-rank characteristic hypothesis. This is necessity for the declared numerical ledger, not intrinsic tightness. The current ePrint also acknowledges the high-rate refinement audited in Section 9.

Provenance. The fixed-multiplicity proof and high-rate certificate construction are recovered from a September 13 research draft. This note incorporates their September 17 local audit, smaller finite certificates, a subsequent uniform finite-length extension, and a converse for the inverse-margin costs of the older bounds, and the restricted current graded-cost converse. No independent referee or formal verification of this note is claimed.

1 The optimization problem and its scope

First-derivative interpolation uses a polynomial in X, Y_0, Y_1 , where a candidate message polynomial P is substituted as $(Y_0, Y_1) = (P, P')$. At an agreement coordinate, writing T for the local coordinate and centering the received value, the expression

$$E = Y_0 - TY_1$$

vanishes to order at least two after substitution. The local multiplicity condition therefore assigns weight one to T and weight two to E .

Fix a positive integer m . Unless explicitly discussing a small-characteristic counterexample, assume that $\mathbb{F}$ has characteristic zero or characteristic greater than every retained jet degree. Define

$$\mathcal{A}_m = \mathbb{F}[T, Y_0, Y_1]/I_m, \quad I_m = (T^i E^j : i + 2j \geq m).$$

The exponent of Y_1 is unrestricted in this ideal. For a finite jet space $V \subset \mathbb{F}[Y_0, Y_1]$, the *saturated local rank* is

$$r_m(V) = \dim_{\mathbb{F}} \text{im}(\mathbb{F}[T]V \longrightarrow \mathcal{A}_m).$$

The image is finite dimensional when the jet degree of V is bounded, because $T^m = 0$ in the quotient.

For a monomial space $\mathbb{F}S$, write its support as

$$S \subset \mathbb{N}^2, \quad (u, b) \leftrightarrow Y_0^u Y_1^b.$$

We require downward closure in u . This is precisely the condition ensuring that the monomial source is stable under translating Y_0 . Let ρ denote the rate and a the agreement fraction. Saturating each allowed jet monomial with the permitted X coefficients gives the leading, block-length-normalized benefit

$$B_{m,\rho,a}(S) = \sum_{(u,b) \in S} (ma - \rho(u + b)). \quad (1)$$

Only positive-benefit monomials need be retained. Thus $u + b < ma/\rho$ throughout.

The object optimized here is

$$B_{m,\rho,a}(S) - r_m(\mathbb{F}S). \quad (2)$$

Positive surplus gives the leading dimension advantage needed for an interpolation certificate. The local rank is the rank of the actual map, not the number of potentially nonzero rows.

Remark 1.1 (What the conclusions do and do not cover). The benefit in (1) is the leading term as block length tends to infinity with the jet space and multiplicity fixed. Exact message-degree weights are D and $D - 1$, not two copies of ρN . Section 6 supplies this bookkeeping for the full-coefficient monomial model, uniformly in multiplicity. The converse below concerns this specified dimension comparison, not every possible interpolation constraint or global source space. In particular, no intrinsic lower bound on exceptional challenges follows from it.

The main discrete conclusion is the following.

Theorem 1.2 (Finite multiplicity at quarter rate). *Put $a_0 = (3 + \sqrt{133})/31$. Under the standing characteristic assumption, for every finite downward- Y_0 support S and every $m \geq 1$, positive surplus at $\rho = 1/4$ and $a < 1/2$ requires*

$$a > a_0 + \frac{1 - 2a_0}{8m}.$$

Conversely, in characteristic zero every $a > a_0$ admits positive surplus for some finite multiplicity and support; the same support works in all characteristics greater than its maximum jet degree. Thus

the infimum threshold in characteristic zero, or over sufficiently large characteristics, is exactly a_0 . The same statements, with the same characteristic quantifiers, hold for the translation-stable jet spaces of Section 5, with benefit defined by their filtered basis degrees.

2 Exact rank and two rearrangements

2.1 The local rank

For a fixed total jet degree q , let

$$S_q = \{b : (q - b, b) \in S\}.$$

Lemma 2.1 (Exact homogeneous rank). *In characteristic zero, or characteristic greater than every jet degree appearing in S ,*

$$r_m(\mathbb{F}S) = \sum_q \sum_{\ell=0}^{m-1} \min(\#\{b \in S_q : q - b \leq \ell\}, m - \ell). \quad (3)$$

Proof. The source column $T^x Y_0^{q-b} Y_1^b$ expands as

$$\sum_{c=b}^q \binom{q-b}{q-c} T^{x+c-b} E^{q-c} Y_1^c.$$

The grade $g = x - b = i - c$ is preserved. In grade g , source columns have $b \geq \max(0, -g)$, and surviving rows have

$$c \geq \max(0, -g, g + 2q - m + 1), \quad c \leq q.$$

Reverse the row suffix, putting $j = q - c$. Its columns are $(\binom{q-b}{j})_j$. Any square minor formed by the first rows and distinct columns is a binomial Vandermonde determinant

$$\frac{\prod_{i < j} (x_j - x_i)}{\prod_{j=0}^{t-1} j!}, \quad x_i = q - b_i,$$

up to sign. The characteristic assumption makes it nonzero. Hence the grade rank is the minimum of the number of columns and rows. Setting $\ell = q + g$ gives row capacity $\min(q + 1, \ell + 1, m - \ell)$. Since the active column count is at most $\min(q + 1, \ell + 1)$, this simplifies to (3). $\square$

Write the support by column heights,

$$S = \{(u, b) : 0 \leq u < H_b\}.$$

Lemma 2.2 (Height sorting). *Replacing (H_b) by its decreasing rearrangement weakly increases benefit and weakly decreases the rank in (3). It preserves the positive-benefit triangle.*

Proof. For a fixed ℓ , clip the heights at $\ell + 1$. The active count on diagonal q is the overlap multiplicity of the integer intervals

$$[b, b + \min(H_b, \ell + 1) - 1].$$

We prove that decreasing rearrangement minimizes the sum of any concave function of these overlaps, normalized to vanish at zero.

Use selection sort from the right. Suppose the fixed suffix is decreasing and no higher than the remaining prefix. Choose a minimum height c in the prefix, move it to its right end, and shift the intervening columns one step left. Split the diagram into the height- c baseline, unchanged earlier excess columns, and the moving excess columns. The baseline is decreasing with height at most c . Its overlap profile is nonincreasing beyond diagonal $c - 1$. Each earlier excess interval also has a nonincreasing profile after the point where the moving excess begins.

Let $C(q)$ be the unchanged profile and $X(q)$ the moving excess after shifting left. On the support of X , $C(q) \geq C(q + 1)$. The new and old costs differ by comparing

$$f(C(q) + X(q)) - f(C(q)) \quad \text{with} \quad f(C(q + 1) + X(q)) - f(C(q + 1)).$$

Concavity makes the first no larger. Summing and repeating the grouped move proves the assertion. Clipping may create equal heights but does not change the argument. Apply it with $f(t) = \min(t, m - \ell)$ and sum over ℓ .

Sorting preserves $\sum H_b$ and $\sum H_b^2$ and decreases $\sum bH_b$, so it increases benefit. The triangle is preserved because a level set of height $u + 1$ originally has at most the number of available columns under that triangle; its rearrangement is the initial interval of that size. $\square$

After sorting, the support is downward in both coordinates. The next rearrangement moves each diagonal toward its smallest second exponent.

Lemma 2.3 (Diagonal compression). *Let k_q be the number of monomials on diagonal q in a two-variable order ideal. Replacing them by $b = 0, \dots, k_q - 1$ preserves the order-ideal property and benefit and weakly decreases rank. After the first incomplete diagonal, k_q is nonincreasing.*

Proof. The complement on an incomplete diagonal has upward shadow containing $C \cup (C + 1)$, with at least $|C| + 1$ elements. Thus the next diagonal has at most as many selected monomials. This proves both the asserted monotonicity and feasibility of the compressed ideal. Moving selected b 's downward moves the corresponding $u = q - b$ upward and can only decrease every active count in (3). $\square$

For a compressed diagonal of length k , the rank marginal has the useful exact form

$$r_{m,q}(k) - r_{m,q}(k - 1) = \max \left(0, m - q, \left\lfloor \frac{m - q + k}{2} \right\rfloor \right). \quad (4)$$

This follows directly by counting the values of ℓ where the minimum in (3) increases. In a compressed column it becomes

$$d_m(u, b) = \max \left(0, m - u - b, \left\lceil \frac{m - u}{2} \right\rceil \right). \quad (5)$$

3 The continuum problem

Put $L = a/\rho$. A height support is a measurable set

$$E_H = \{(x, s) : s \geq 0, 0 \leq x < H(s)\}, \quad 0 \leq H(s) \leq (L - s)_+.$$

Its benefit and exact continuum rank are

$$\begin{aligned}\mathcal{B}_{\rho,a}(E) &= \int_E (a - \rho(x+s)) dx ds, \\ \mathcal{R}(E) &= \int_{\mathbb{R}} \int_0^1 \min(A_E(h,t), 1-t) dt dh, \\ A_E(h,t) &= \int_0^t 1_E(x, h-x) dx.\end{aligned}$$

Write $\mathcal{P}_{\rho,a} = \mathcal{B}_{\rho,a} - \mathcal{R}$.

Lemma 3.1 (Continuum compression). *Every height support is dominated in surplus by one of the form*

$$E = \{(x, s) : 0 \leq s \leq B, s \leq x + s \leq R(s)\}, \quad (6)$$

where R is nonincreasing and $s \leq R(s) \leq L$.

Proof. The height-sorting proof applies directly to equal-width step heights. At a fixed clipping height, the diagonal profile of an excess rectangle is a convolution of interval indicators. The unchanged earlier rectangles have stopped increasing before the shifted excess begins. The same concavity comparison therefore applies. For general heights, approximate in L^1 . Decreasing rearrangement contracts the L^1 distance, and

$$\int_h |A_{E_H}(h, t) - A_{E_G}(h, t)| dh \leq \|H - G\|_1.$$

Thus rank passes to the limit. The benefit comparison follows by layer cake: initial intervals minimize the first moment of a set of prescribed measure. The triangle condition follows from $|\{H > x\}| \leq (L - x)_+$.

The resulting support is downward in both coordinates; choose its canonical monotone representative. On a diagonal h , let Q_h be its complement in $[0, h]$. For $\delta > 0$,

$$Q_h + [0, \delta] \subseteq Q_{h+\delta}.$$

Once the complement has positive measure, its one-dimensional measure grows by at least δ ; the transition follows by a limit. Consequently the selected diagonal length is initially h and then is nonincreasing. Replacing each diagonal by its initial interval in s preserves these lengths and benefit, and moves selected $x = h - s$ upward, decreasing $A_E(h, t)$ pointwise. The compressed set has exactly the form (6). Null-set boundary choices are immaterial. $\square$

For a compressed diagonal, differentiating the rank with respect to its length gives

$$f(h, s) = \max\left(0, 1 - h, \frac{1 - h + s}{2}\right).$$

It follows that the surplus of (6) is

$$\int_0^B F_s(R(s)) ds, \quad F_s(R) = \int_s^R w_s(h) dh, \quad w_s(h) = a - \rho h - f(h, s). \quad (7)$$

The same formula follows as the scaled limit of (4).

3.1 Rates at most one half

Lemma 3.2 (Prefix completion). *Suppose $F_s(R) \leq 0$ for all $s \geq L - 1$ and $s \leq R \leq L$. Then a full-column cap, with $R(s) = L$ on a prefix, maximizes surplus.*

Proof. Delete columns $s \geq L - 1$. For $s < L - 1$, the density w_s has a single-crossing property: once positive, it remains positive up to L . On the first rank branch it increases with slope $1 - \rho$; on the middle branch a negative slope is possible, but its right-end value is $a - \rho(1 + s) > 0$; after that it equals the positive benefit. Also $w_s(h)$ is nonincreasing in s at fixed h .

Take a maximizer among decreasing endpoints; compactness follows from Helly's theorem and bounded convergence. If an unsaturated endpoint has positive density, completing its preceding prefix to L improves the integral: every earlier endpoint is at least as large and has at least as large a density there, and single crossing controls the whole added interval. Applying this argument at density points excludes a positive-measure set of such endpoints. Hence almost every unsaturated column ends where $w_s \leq 0$ and has nonpositive total surplus. These columns form a suffix, which can be deleted. What remains is a cap. $\square$

If $\rho \leq 1/2$ and $s \geq L - 1$, then for $h \leq L$,

$$w_s(h) \leq a - \frac{1+s}{2} + (1/2 - \rho)h \leq \frac{L-1-s}{2} \leq 0.$$

Thus caps are globally optimal for every $0 < \rho \leq 1/2$ and $\rho < a < \sqrt{\rho}$.

3.2 The critical quarter-rate cap

Lemma 3.3 (Quarter-rate continuum threshold). *At rate $1/4$, every admissible height support has nonpositive surplus when $a = a_0 = (3 + \sqrt{133})/31$. Every $a > a_0$ admits a support with positive surplus.*

Proof. The preceding compression and prefix-completion lemmas reduce the problem to a full cap. For width $B \leq 1/2$, integration of the rank marginal gives cap surplus

$$P_a(B) = (2a^2 - 1/2)B + \frac{1-a}{2}B^2 - \frac{7}{24}B^3.$$

The positive root of $31a^2 - 6a - 4 = 0$ is a_0 . At that value,

$$P_{a_0}(B) = -\frac{7}{24}B \left(B - \frac{6(1-a_0)}{7} \right)^2 \leq 0.$$

The nonempty zero occurs at a width strictly below $1/2$ and below $4a_0 - 1$. For $s \geq 1/2$, the additional cap column has surplus $2(a_0 - s/4)^2 - 1/4$, which is negative throughout the relevant range $1/2 \leq s \leq 4a_0 - 1$. Indeed it is already negative at $s = 1/2$ and then decreases. Columns beyond $4a_0 - 1$ can be removed by the preceding lemma. Thus no larger cap improves the value. Increasing a strictly increases surplus on the nonempty zero cap, proving the second claim. $\square$

4 A one-sided finite-multiplicity comparison

We now prove Theorem 1.2. This step is essential: a continuum limit with an unsigned error does not exclude an exceptional finite multiplicity.

Use unscaled continuum coordinates and set

$$f_m(h, s) = \max\left(0, m - h, \frac{m - h + s}{2}\right), \quad w_p(h, s) = p - h/4 - f_m(h, s).$$

For integers $0 \leq b \leq Q \leq 2m - 1$, define

$$F_{p,b}(Q) = \sum_{q=b}^Q \left[p - \frac{q}{4} - \max\left(0, m - q, \left\lceil \frac{m - q + b}{2} \right\rceil \right) \right].$$

Lemma 4.1 (One-sided column lift). *For $p \leq m/2$,*

$$\int_b^{b+1} \int_s^{Q+1} w_p(h, s) dh ds \geq F_{p,b}(Q) + \frac{m - 2p}{4}. \quad (8)$$

Proof. Let $n = Q - b + 1$ and denote the integral minus the sum by $D_p(Q)$. Their areas differ by $1/2$, so

$$D_p(Q) = D_{m/2}(Q) + (m - 2p)/4.$$

It suffices to prove $D_{m/2}(Q) \geq 0$.

Let $I_{q,b}$ be the integral of $f_m(q + v, b + y)$ over $0 \leq v, y \leq 1$, and let $\delta_{q,b}$ be its difference from the discrete rank marginal, in the order discrete minus integral. Direct integration gives

Range	$\delta_{q,b}$
$q \leq m - b - 2$	$1/2$
$q = m - b - 1$	$5/12$
$m - b \leq q \leq m + b - 1$	0 if $m - q + b$ is even, $1/2$ if odd
$q = m + b$	$-1/12$
$q \geq m + b + 1$	0

Only rows with $q \geq b$ are used. The two exceptional corrections follow from integrating $(v + y - 1)_+/2$ and $(y - v)_+/2$ over the square; each integral is $1/12$.

The desired column is the union of the n squares with its bottom triangle $s > h$ removed. On that triangle,

$$f_m(b + v, b + y) \geq (m + y - v)/2,$$

so the removed rank is at least $m/4 + 1/12$. Each full square loses $1/8$ of benefit relative to its lattice point, while the removed triangle has benefit $p/2 - b/8 - 1/24$. Therefore

$$D_{m/2}(Q) \geq \frac{b+1}{8} + \sum_{q=b}^Q (\delta_{q,b} - 1/8). \quad (9)$$

If $Q \leq m + b - 1$, the summands consist of initial positive terms and an alternating sequence $-1/8, 3/8$, starting at either parity. Every partial sum is at least $-1/8$, proving nonnegativity.

At and after $q = m + b$, the right side of (9) decreases with Q , so it suffices to take $Q = 2m - 1$. This case requires $b < m$. For $b \geq 1$, compare the first m terms, from $q = b$ through $m + b - 1$, with their central parity sequence. All dominate it except possibly the $5/12$ term, which loses at most $1/12$. The boundary term loses another $1/12$. Hence

$$\sum_{q=b}^{2m-1} \delta_{q,b} \geq \frac{1}{2} \left\lceil \frac{m}{2} \right\rceil - \frac{1}{6} \geq \frac{m}{4} - \frac{1}{6}.$$

Substitution into (9) gives $D_{m/2} \geq b/4 - 1/24 > 0$. If $b = 0$, the sum is exactly $m/2 - 1/6$, giving $D_{m/2} \geq m/4 - 1/24 > 0$. This proves the lemma. $\square$

Proof of Theorem 1.2. Apply height sorting and diagonal compression. The resulting nonempty support has C columns

$$b = 0, \dots, C-1, \quad q = b, \dots, Q_b, \quad Q_0 \geq Q_1 \geq \dots \geq Q_{C-1}.$$

All $Q_b < 2m$ when $a < 1/2$. Lift it to columns $b \leq s < b+1$, $s \leq h < Q_b + 1$. Their endpoints are decreasing, so the integral of w_{ma} is the actual continuum surplus. Lemma 4.1 shows that this surplus is at least

$$B_{m,1/4,a}(S) - r_m(\mathbb{F}S) + \frac{Cm(1-2a)}{4}.$$

The lift may protrude beyond $h = 4ma$. Delete that part: its density is nonpositive, and the remaining endpoints are still decreasing. Rescaling both coordinates by m gives an admissible support in the continuum problem.

At $a = a_0$, Theorem 3.3 gives nonpositive continuum surplus. In fact the critical cap polynomial is

$$-\frac{7}{24}B \left(B - \frac{6(1-a_0)}{7} \right)^2$$

for $B \leq 1/2$, and decreases thereafter. Consequently

$$B_{m,1/4,a_0}(S) - r_m(\mathbb{F}S) \leq -\frac{Cm(1-2a_0)}{4}. \quad (10)$$

The same conclusion applies to any compressed support with $Q_b < 2m$, even if some of its benefits at a_0 are negative: the truncation still improves surplus.

Monotonicity first excludes $a \leq a_0$. For $a > a_0$, use $|S| \leq 2mC$ and (10) to obtain

$$B_{m,1/4,a}(S) - r_m(\mathbb{F}S) \leq -\frac{Cm(1-2a_0)}{4} + 2m^2C(a-a_0).$$

This proves the necessary strict inequality. Conversely, a cap has strictly positive continuum surplus for $a > a_0$. Approximating it on sufficiently fine integer grids gives positive finite surplus in characteristic zero, and then in every characteristic exceeding the resulting maximum jet degree. Here an ordinary two-sided approximation error is sufficient, because the limiting margin is positive. No approximation claim is made within one fixed positive characteristic under a fixed jet-degree guard. $\square$

5 Nonmonomial translation-stable sources

The monomial restriction can be removed for a natural source class. Let

$$V \subseteq \mathbb{F}[Y_0, Y_1]_{\leq d}$$

be finite dimensional and stable under Y_0 -translation. Define its filtered degree multiplicities by

$$n_q = \dim(V \cap \mathbb{F}[Y_0, Y_1]_{\leq q}) - \dim(V \cap \mathbb{F}[Y_0, Y_1]_{\leq q-1}),$$

and its benefit by

$$B_{m,\rho,a}(V) = \sum_q n_q(ma - \rho q)_+.$$

Theorem 5.1 (Monomial domination). *Suppose $\text{char } \mathbb{F} = 0$ or $\text{char } \mathbb{F} > d$. There is a downward- Y_0 monomial support S , independent of m , such that*

$$|S_q| = n_q, \quad r_m(\mathbb{F}S) \leq r_m(V) \quad (m \geq 1).$$

Thus benefit is preserved and every local saturated rank weakly decreases. The quarter-rate finite theorem and the continuum conclusions transfer to this source class.

Proof. The jet-degree-at-most- d part of $\mathcal{A}_m$ is finite dimensional, with basis

$$T^i E^j Y_1^\ell, \quad i + 2j < m, \quad j + \ell \leq d.$$

Since $T^m = 0$, the saturated rank is already the rank of the image of $\bigoplus_{i=0}^{m-1} T^i V$ in this fixed finite target. Matrix rank cannot increase at specialization.

Choose a filtration-adapted basis v_i with degrees q_i . The family

$$t^{q_i} v_i(t^{-1}Y_0, t^{-1}Y_1)$$

has linearly independent highest parts at $t = 0$ and degenerates V to its associated graded space. Generically the ambient scaling preserves I_m , sending E to $t^{-1}E$, so the local rank is unchanged. The limiting rank is no larger, and degree multiplicities are preserved.

Now act on this homogeneous space by $Y_1 \mapsto tY_1$. Extend the action to the local target by $T \mapsto t^{-1}T$ and $Y_0 \mapsto Y_0$. It fixes E and preserves I_m . Saturation in T makes the generic local rank invariant. In each homogeneous degree, row reduction in increasing Y_1 exponent gives distinct pivot monomials; rescaling the corresponding basis vectors yields a monomial limiting space. Again rank cannot increase, and all degree multiplicities remain unchanged. Neither limiting source depends on m .

Finally, the finite Taylor formula shows that ∂_{Y_0} -stability implies translation stability. Conversely, coefficient extraction at $d + 1$ distinct shifts, available under the characteristic assumption, gives derivative stability. Derivative stability is a closed condition on the Grassmannian, so both limits preserve it. For a monomial $Y_0^u Y_1^b$ with $u > 0$, its derivative has nonzero coefficient u ; hence the limiting support contains $(u - 1, b)$. This proves downward closure.

Discarding nonpositive-benefit monomials preserves downward closure, leaves benefit unchanged, and can only decrease rank. The conclusions now follow from the monomial results. $\square$

The characteristic assumption has content. In characteristic two, $V = \text{span}\{1, Y_0^2\}$ is translation-stable and has $r_2(V) = 2$, because $I_2 = (T^2, E)$ makes $Y_0^2 = 0$. A downward- Y_0 monomial support with the same filtered degrees $0, 2$ must be $\{1, Y_1^2\}$, whose rank is four.

The theorem concerns fixed jet spaces, their total-degree filtration, and saturated local sources. It does not assert that arbitrary global interpolation spaces or alternative local ideals degenerate to this model.

6 Exact finite-length weights, uniformly in multiplicity

Write $a_0 = (3 + \sqrt{133})/31$ and $c_0 = (1 - 2a_0)/8$. The following result does not keep multiplicity fixed as length grows.

Theorem 6.1 (Uniform finite-length converse). *Let $N \geq 12$ be divisible by four, $k = N/4$, $D = k - 1$, and $k < A \leq N$ be an integer. Let $m \geq 1$ be arbitrary. For a finite downward- Y_0 monomial jet support, retain all coefficient monomials*

$$X^x Y_0^u Y_1^b, \quad x + Du + (D - 1)b < mA.$$

Discard jets with no permitted coefficients. Let G be the resulting source dimension and R its exact local rank modulo I_m . Assume characteristic zero or characteristic greater than every retained total jet degree. If the sufficient dimension count $G > NR$ holds, then

$$\frac{A}{N} > \left(1 - \frac{8}{N}\right) \left(a_0 + \frac{c_0}{m}\right) + \frac{8}{N^2}. \quad (11)$$

In particular, uniformly over all choices $m = m(N)$, the asymptotic agreement threshold is at least a_0 .

We first record the exact rank before coefficient saturation. Put

$$L_q = mA - (D - 1)q, \quad U_q = \{u : (u, q - u) \in S, u < L_q\}.$$

On total jet degree q , the source dimension and local rank are

$$G_q = \sum_{u \in U_q} (L_q - u), \quad (12)$$

$$R_q = \sum_{\ell=0}^{\min(m, L_q)-1} \min(\#\{u \in U_q : u \leq \ell\}, m - \ell). \quad (13)$$

The sum is empty when $L_q \leq 0$. To see the second formula, use the grade $g = x - b$ from Lemma 2.1. The coefficient cutoff becomes $g < mA - Dq$, independently of b . With $\ell = q + g$ this is exactly $\ell < L_q$. The same binomial Vandermonde argument therefore applies. The active column count already absorbs the row bounds $q + 1$ and $\ell + 1$. Translation in X preserves the full coefficient prefixes; translation in Y_0 preserves the source and lowers weighted degree. Consequently this local rank is independent of the coordinate and received value.

Lemma 6.2 (Half-rank inequality). *For $L \geq 1$ and $U \subseteq \{0, \dots, L - 1\}$, define*

$$G_U = \sum_{u \in U} (L - u), \quad R_U = \sum_{\ell=0}^{L-1} \min(\#\{u \in U : u \leq \ell\}, L - \ell).$$

Then $G_U \leq 2R_U$.

Proof. Order U as $u_1 < \dots < u_t$. The source cells are pairs (j, ℓ) with $u_j \leq \ell < L$; retain a cell when $j \leq L - \ell$. Send every discarded cell injectively to

$$(j', \ell') = (L - \ell, L - j).$$

Since $1 \leq j' < j \leq t \leq L$, both indices are valid. Distinctness of the ordered integers gives $u_{j'} \leq u_j - (j - j') \leq \ell - j + j' = L - j = \ell'$. Moreover $j' \leq j = L - \ell'$, so the image is retained. There are at most as many discarded cells as retained cells. $\square$

Proof of the uniform finite-length converse. For $0 < L_q < m$, formula (13) is at least R_{U_q} with $L = L_q$. Lemma 6.2 gives $G_q \leq 2R_q$, so this diagonal contributes nonpositively to $G - NR$. Remove all diagonals with $L_q < m$. As $D - 1 > 0$, these form a final segment of total degrees; the surviving support S_0 remains downward in Y_0 . Its local rank is now the saturated rank $r_m(\mathbb{F}S_0)$, and its source dimension satisfies $G_0 > Nr_m(\mathbb{F}S_0)$.

Every surviving degree satisfies $q \leq m(A - 1)/(D - 1)$. Writing $a = A/N$, the normalized coefficient count of a retained jet is

$$\frac{mA - Dq + b}{N} = ma - \frac{q}{4} + \frac{q + b}{N} \leq ma_* - \frac{q}{4},$$

where

$$a_* = \frac{A}{N} + \frac{2(A - 1)}{N(D - 1)} = \frac{AN - 8}{N(N - 8)}.$$

Thus the saturated quarter-rate comparison has positive surplus at a_* . All its retained benefits are positive. If $a_* < 1/2$, Theorem 1.2 gives $a_* > a_0 + c_0/m$. If $a_* \geq 1/2$, the same strict inequality follows from $a_0 + c_0 < 1/2$. Rearranging yields (11). $\square$

The bound implies $A > Na_0 - 8a_0 + 8/N$, so the possible finite-length improvement is bounded by a constant number of coordinates. The proof uses the saturated theorem separately for each integer m ; it takes no limit with m fixed. Its scope is the displayed monomial source with full coefficient prefixes. It does not establish a finite-length nonmonomial degeneration or rule out kernels caused by dependencies among constraints at different coordinates.

7 Inverse-margin costs of the older reconstruction bounds

The quarter-rate converse also controls the dependence on the margin above the threshold. Throughout this section, benefits are the leading normalized quantities in (1). The conclusion concerns the specific interpolation and reconstruction bounds in the recovered September 10 working draft [1], not the smallest possible actual lists. The public ePrint [2] replaces those reconstruction formulas and proves better inverse-margin powers. Its graded challenge counts, Equations (63)–(64), also go beyond the uniform challenge count analyzed here.

Theorem 7.1 (Necessary parameter costs). *Let $a = a_0 + \varepsilon$, where $0 < \varepsilon \leq 1/2000$, and put $c_0 = (1 - 2a_0)/8$. Suppose a nonempty downward- Y_0 monomial support S has positive benefits and positive surplus at rate $1/4$. Write*

$$B = B_{m,1/4,a}(S), \quad R = r_m(\mathbb{F}S), \quad \Delta = B - R, \quad s = |S|,$$

and let $U = \max_S u$, $V = \max_S b$, and $J = \max_S (u + b)$. Then

$$m > \frac{c_0}{\varepsilon}, \quad U > \frac{m}{4}, \quad V > \frac{m}{4} - 1, \quad \frac{R}{\Delta} \geq \frac{1}{8\varepsilon}. \quad (14)$$

In particular, $m \geq 16$ and $V \geq m/8$.

Proof. The multiplicity bound is Theorem 1.2. Direct substitution in $31a^2 - 6a - 4$ bounds $a_0 < 469/1000$, so $a < 47/100$ and $c_0 > 3/400$. Thus $m > 15$.

If $U \geq m/2$, its claimed bound is immediate. Otherwise each total-degree diagonal has at most $U + 1$ monomials, all active in (3) for $\ell \geq U$. For $U \leq \ell \leq m - U - 1$, row capacity is at least $U + 1$. Consequently $R \geq (m - 2U)s$. Since $B \leq mas$ and $B > R$, we obtain $U > m(1 - a)/2 > m/4$.

Apply height sorting and diagonal compression. Neither increases the maximum second exponent; the compressed support therefore has at most $V + 1$ columns. Its one-sided continuum lift has strictly positive surplus by Lemma 4.1. Trimming outside the benefit triangle improves surplus. Continuum compression and prefix completion can be performed within the existing width: sorting replaces level sets by initial intervals of the same measure, diagonal compression never increases

their maximal second coordinate, and completion only adds to existing columns. Thus there is a positive cap of width at most $(V + 1)/m$.

For $0 < w \leq 1/4$ and $a_0 \leq a \leq 47/100$, the cap surplus divided by width is

$$2a^2 - \frac{1}{2} + \frac{1-a}{2}w - \frac{7}{24}w^2.$$

Both partial derivatives are positive on this rectangle, and its value at $(a, w) = (47/100, 1/4)$ is $-2443/240000$. Hence positive surplus requires width greater than $1/4$, proving $V > m/4 - 1$. Since $m \geq 16$, this implies $V \geq m/8$.

For the ratio bound, rearrange again. Every compressed column has height $h \leq 2m$, since positive benefit implies total degree less than $4ma < 2m$. By (5), its rank is at least $\sum_{u=0}^{h-1} (m-u)_+/2$. For $h \leq m$ this is at least $mh/4$; for $m \leq h \leq 2m$ it is at least $m(m+1)/4 \geq mh/8$. Rearrangement preserves cardinality and decreases rank, so the original support satisfies $R \geq ms/8$.

At a_0 , every support has nonpositive surplus. This remains true when some benefits are nonpositive: deleting those monomials preserves downward closure, increases benefit, and decreases rank. Therefore

$$\Delta \leq B_{m,1/4,a}(S) - B_{m,1/4,a_0}(S) = m\epsilon s.$$

Combining the last two inequalities proves the ratio bound. $\square$

Corollary 7.2 (Translation-stable nonmonomial spaces). *Let $W \subseteq \mathbb{F}[Y_0, Y_1]_{\leq d}$ be translation-stable in Y_0 , in characteristic zero or greater than d . Give it the filtered positive-part benefit $B = B_{m,1/4,a}(W)$ of Section 5. Suppose $\Delta = B - r_m(W) > 0$. If every polynomial in W has Y_0 degree at most U and Y_1 degree at most V , all four inequalities in (14) hold with $R = r_m(W)$.*

Proof. The two flat degenerations in Theorem 5.1 preserve the ambient rectangle of monomials $u \leq U$, $b \leq V$: coordinate scalings preserve that rectangle, and taking leading parts or limits cannot create a monomial outside it. They also preserve the total-degree filtration. Let S be their downward- Y_0 monomial limit, and delete monomials with nonpositive benefit. The retained support S_+ still has downward closure, has benefit exactly B , and has rank $R_+ \leq r_m(W) = R$. It is nonempty and has positive surplus $B - R_+ \geq \Delta > 0$. Apply Theorem 7.1 to S_+ . Its coordinate degrees are at most U, V , giving the first three inequalities. For the fourth, since $0 \leq R_+ \leq R < B$,

$$\frac{R}{B - R} \geq \frac{R_+}{B - R_+} \geq \frac{1}{8\epsilon}.$$

$\square$

Corollary 7.3 (Sharp margin powers for the older stated bounds). *In either of the preceding source classes, suppose the uniform challenge-degree budget ℓ is certified by*

$$(\ell + 1)B > (\ell + \beta + 1)R, \quad \beta \geq U.$$

Then $\ell = \Omega(\epsilon^{-2})$. Suppose further that the reconstruction parameters satisfy $B_\partial \geq V$ and $B_{\text{jet}} \geq \max(J, B_\partial)$, and use

$$B_{\text{rec}} = (B_{\text{jet}} - B_\partial)B_\partial(B_\partial + 1) + \frac{B_\partial(B_\partial + 1)(2B_\partial + 1)}{6}.$$

The unchanged list and MCA budgets of [1], Theorem 1.1 and Equation (23), have orders at least

$$\Omega(D\epsilon^{-3}) \quad \text{and} \quad \Omega(D^2\epsilon^{-5}),$$

respectively, where D is the message-degree cap. At quarter rate these match the margin powers in that draft's upper estimates.

Proof. The integer challenge budget obeys

$$\ell \geq \left\lfloor \frac{\beta R}{\Delta} \right\rfloor > \frac{m}{32\varepsilon} - 1 > \frac{c_0}{32\varepsilon^2} - 1.$$

Also $B_{\text{rec}} \geq B_{\partial}^3/3 = \Omega(\varepsilon^{-3})$. Writing $\lambda = (n - D)/(A - D) \geq 1$, the list budget contains $2D\lambda B_{\text{rec}}$, and the MCA budget contains $24D^2\ell\lambda^2 B_{\text{rec}}$. All their other terms are nonnegative. These terms imply the asserted orders. $\square$

These are sharp powers for these particular sufficient bounds, even after arbitrary support optimization or passage to translation-stable nonmonomial jet spaces with the stated filtered benefit. They are not lower bounds on actual kernel-vector degrees, candidate lists, or exceptional challenges. Global constraint dependencies, finer coefficient-degree accounting, or different reconstruction and agreement arguments could improve the bounds. The statements above use leading coefficients. For the monomial source with full coefficient prefixes, the following refinement gives exact finite-length accounting.

Corollary 7.4 (Exact finite-length parameter costs). *Use the source, degree bounds U, V , dimension G , and local rank R of Section 6. Suppose $\Delta = G/N - R > 0$. Put*

$$a_* = \frac{AN - 8}{N(N - 8)}, \quad \varepsilon_* = a_* - a_0.$$

If $0 < \varepsilon_ \leq 1/2000$, all four inequalities in (14) hold with ε_* in place of ε . Thus the exact uniform count*

$$(\ell + 1)G > N(\ell + \beta + 1)R, \quad \beta \geq U,$$

requires $\ell = \Omega(\varepsilon_^{-2})$. The unchanged reconstruction budgets have orders at least $\Omega(D\varepsilon_*^{-3})$ and $\Omega(D^2\varepsilon_*^{-5})$. These assertions are uniform in m .*

Proof. The finite-length reduction deletes final diagonals with nonpositive surplus and produces a downward support S_0 of saturated rank R_0 . Writing G_0 for its coefficient dimension and B_* for its leading benefit at a_* , that reduction proves

$$R_0 \leq R, \quad \Delta_0 = G_0/N - R_0 \geq \Delta > 0, \quad B_* \geq G_0/N.$$

All retained benefits at a_* are positive. Applying Theorem 7.1 to S_0 proves the multiplicity and coordinate degree bounds, since its degrees do not exceed those of the original source. Moreover,

$$\frac{R}{\Delta} \geq \frac{R_0}{\Delta_0} \geq \frac{R_0}{B_* - R_0} \geq \frac{1}{8\varepsilon_*}.$$

The challenge and reconstruction calculations are unchanged. $\square$

If $A/N = a_0 + \varepsilon$, the exact effective margin is

$$\varepsilon_* = \frac{N\varepsilon + 8a_0 - 8/N}{N - 8}.$$

For $\varepsilon \geq 1/N$, it is $\Theta(\varepsilon)$ as N grows. At the rounded threshold $A = \lceil Na_0 \rceil$, it satisfies $0 < \varepsilon_* < 5/(N - 8)$. Consequently this framework requires $m = \Omega(N)$ and $\ell = \Omega(N^2)$ there, and its displayed list and MCA budgets are respectively $\Omega(N^4)$ and $\Omega(N^7)$. These remain costs of the specified sufficient bounds, not lower bounds on actual lists or exceptional sets. The exact finite-length extension here is monomial; no finite-length nonmonomial degeneration is asserted.

8 Necessary costs of the current order-ideal bounds

We now use the current ePrint's graded challenge count and squarefree reconstruction formulas [2, Equations (54)–(58), (60), (63)–(64)]. For monomial supports with full derivative-weighted coefficient prefixes, their quadratic list-budget and fourth-power regular-MCA-budget dependence cannot be improved by choosing a different such support or a smaller height certified by the graded row test on coordinatewise order ideals, or by the column-only test on supports downward in Y_0 . This is a statement about these numerical bounds, not actual component degrees, lists or exceptions. The support restriction below is essential.

Theorem 8.1 (Current order-ideal costs). *Let $N \geq 12$ be divisible by four, $D = N/4 - 1$, and let $D + 1 < A \leq N$ be an integer. Let S be a nonempty finite monomial support downward in Y_0 , let $m \geq 1$, $B = \max_S(u + v)$ and $b = \max_S v$, and use all monomials*

$$X^x Y_0^u Y_1^v, \quad (u, v) \in S, \quad x + Du + (D - 1)v < mA, \quad x, u, v \geq 0.$$

Assume characteristic zero or greater than every retained total jet degree. Write G_q, R_q for the exact source dimension and local rank on total jet degree q , and put

$$a_* = \frac{AN - 8}{N(N - 8)}, \quad \varepsilon_* = a_* - a_0, \quad c_0 = \frac{1 - 2a_0}{8}.$$

Suppose $0 < \varepsilon_ \leq 1/2000$. If S is also downward in Y_1 and an integer $H \geq 0$ passes the following graded row test for an affine received line:*

$$\sum_q (H - q + 1)_+ (G_q - NR_q) > 0. \quad (15)$$

Then

$$m > \frac{c_0}{\varepsilon_*}, \quad B > \frac{m}{4}, \quad b \geq \frac{m}{8}, \quad H + 1 > \frac{m}{36864\varepsilon_*}, \quad H > \frac{m}{73728\varepsilon_*}. \quad (16)$$

Without requiring downward closure in Y_1 , if $G_u^{(0)}$ counts source coefficients of Y_0 exponent u and $G = \sum_u G_u^{(0)}$, $R = \sum_q R_q$, the column test

$$\sum_u (H - u + 1)_+ G_u^{(0)} > NR(H + 1)$$

gives the same bounds on m, B, b , and

$$H + 1 > \frac{m}{589824\varepsilon_*}, \quad H > \frac{m}{1179648\varepsilon_*}.$$

Under either test, the current squarefree list budget is therefore $\Omega(D\varepsilon_^{-2})$, and its regular-family MCA budget is $\Omega(D^2\varepsilon_*^{-4})$, even after optimizing the latter's integer incidence threshold.*

Proof. Normalize the prefix surpluses by N :

$$P_j = \sum_{q \leq j} (G_q/N - R_q).$$

The left side of (15), divided by N , is $\sum_{j=0}^H P_j$. Some prefix is therefore positive. Total-degree truncation preserves the full coefficient prefixes and downward- Y_0 closure. Applying the exact

finite-length parameter-cost corollary of Section 7 to that prefix gives $m > c_0/\varepsilon_*$, maximum Y_0 degree greater than $m/4$, and maximum Y_1 degree greater than $m/4 - 1$. Hence $m \geq 16$, $B > m/4$, $b \geq m/8$, and $H > m/4 - 1 \geq m/8$.

For every prefix, whether positive or negative, the reduction in Section 6 deletes final diagonals of nonpositive surplus and produces a saturated support S_0 . Its exact coefficient benefit is dominated by the leading benefit at a_* , with every retained benefit positive. Thus each retained total degree is less than $4ma_* < 2m$, and $|S_0| \leq 2m^2 + m \leq 3m^2$. The critical support converse at a_0 yields

$$P_j \leq B_{m,1/4,a_*}(S_0) - r_m(S_0) \leq m\varepsilon_*|S_0| \leq 3\varepsilon_*m^3. \quad (17)$$

The middle inequality remains valid if some benefits at a_0 are nonpositive, by deleting those monomials. Empty S_0 causes no problem.

To force a triangle in the original support, apply the preceding finite reduction to a positive prefix. Its retained order ideal has positive leading surplus. If d is its first incomplete diagonal, every diagonal has length at most d , by the upward-shadow argument in Lemma 2.3. Diagonal compression preserves leading benefit and weakly decreases rank, and has derivative degree at most $d - 1$. The positive-surplus derivative-degree bound therefore gives $d > m/4$. Consequently, putting $J = \lfloor m/8 \rfloor$, the original support contains all $q + 1$ jet monomials on every degree $q \leq J$. Their coefficient prefixes are nonempty and locally saturated, since

$$L_q = mA - (D - 1)q \geq m(A - (D - 1)/8) \geq m.$$

The exact rank formula, restricted to $\ell = q, \dots, m - q - 1$, gives

$$R_q = \sum_{\ell=0}^{m-1} \min\{\min(q, \ell) + 1, m - \ell\} \geq (m - 2q)(q + 1) \geq \frac{3m(q + 1)}{4}.$$

The exact coefficient comparison gives $G_q/N \leq ma_*(q + 1)$. As $a_* < 47/100$, we obtain

$$G_q/N - R_q \leq -\frac{m(q + 1)}{4}, \quad P_j \leq -\frac{m(j + 1)(j + 2)}{8} \quad (0 \leq j \leq J).$$

Since $H \geq J$, the negative prefix area in the graded sum is at least

$$-\sum_{j=0}^J P_j \geq \frac{m(J + 1)(J + 2)(J + 3)}{24} > \frac{m^4}{12288}.$$

Every positive prefix is at most $3\varepsilon_*m^3$ by (17). Positivity of the entire sum implies $3\varepsilon_*m^3(H + 1) > m^4/12288$, proving the $H + 1$ bound. Since $H \geq m/8 \geq 2$, also $H \geq (H + 1)/2$, proving the last inequality in (16).

For the column test, first reduce a merely downward- Y_0 support to an order ideal without increasing H, B, b . Write $h = H + 1$ and $w_u = (h - u)_+$. A final diagonal with $L_q < m$ contributes at most $h(G_q - NR_q) \leq 0$ to the column margin, by the finite half-rank lemma. Delete these diagonals. The retained local rank is saturated, and the retained jet support still has column heights h_v . Its exact weighted coefficient contribution in column v is

$$\alpha(h_v) - (D - 1)v\beta(h_v), \quad \alpha(d) = \sum_{u < d} w_u(mA - Du), \quad \beta(d) = \sum_{u < d} w_u.$$

Since β is nondecreasing, sorting the heights in decreasing order increases this coefficient sum by rearrangement. The height-sorting lemma weakly decreases the saturated local rank. Sorting

respects every decreasing upper bound on column heights: otherwise the first violating sorted height would require more large original columns than the bound allows. In particular it preserves the total-degree cutoff used for saturation, the original total-degree cap B , and positive coefficient prefixes; the derivative cap b cannot increase. Thus the same column test holds for the sorted order ideal, with no larger declared caps.

For this order ideal, put $\Delta = G/N - R$. The column test is exactly

$$(H+1)\Delta > \sum_u \min(u, H+1)G_u^{(0)}/N.$$

Thus $\Delta > 0$; the same mandatory-triangle argument applies to the full support, and $\Delta \leq 3\varepsilon_* m^3$. The support truncated to $u \leq H$ also has positive surplus: its source dimension exceeds NR , and its rank is at most R . The finite converse gives $H > m/4$. Every triangle monomial has at least $Nm/8$ coefficient monomials, since $(mA - Du - (D-1)v)/N \geq (mA - Dm/8)/N \geq m/8$. Hence, using $J \geq m/16$,

$$\sum_u \min(u, H+1)G_u^{(0)}/N \geq \frac{m}{8} \frac{J(J+1)(J+2)}{6} \geq \frac{m^4}{196608}.$$

This proves the two stated column-test height bounds.

For the current reconstruction formulas, write

$$\tau = 2D - 3, \quad u = 1 + \tau(B-1), \quad v = \min\{u, \tau(b-1) + D\}, \quad F = Bv + b(u-v).$$

We have $D \geq 2$, $B \geq 2$, $0 \leq v \leq u$ and $\tau \geq D/2$. Thus

$$F = bu + (B-b)v \geq bu \geq \frac{DBb}{4} > \frac{Dm^2}{128}.$$

The squarefree list budget is $\lambda_{\text{agr}}F + S$, where $\lambda_{\text{agr}} = (N-D)/(A-D) \geq 1$ and $S \geq 0$. It is at least $Dc_0^2/(128\varepsilon_*^2)$. The declared joint-degree bound is

$$J_{\text{reg}} = H(2uv - v^2) + 2(1 + \tau H)F \geq \frac{D^2HBb}{4} > \frac{D^2c_0^3}{150994944\varepsilon_*^4}.$$

For every permitted incidence threshold $D+1 \leq L \leq A$, its multiplier in the current regular MCA bound is

$$\frac{N-L+1}{A-L+1}\lambda_{\text{agr}} \geq 1.$$

The other terms are nonnegative, completing the proof. $\square$

Remark 8.2 (Scope of current-method tightness). The mandatory triangle follows for every coordinatewise order ideal, without assuming that rearrangement preserves a graded moment. For the row test, supports downward only in Y_0 remain outside the argument; for the column test, height sorting handles this larger class. Neither argument covers nonmonomial spaces or the alternative equal-weight coefficient cutoff. The characteristic hypothesis remains the stronger local-rank hypothesis of this note, not merely the reconstruction condition $p > \max\{D, b\}$. Optimizing actual component degrees, exploiting global kernel dependencies, or using additional equations is outside the numerical ledger considered here. When $A/N = a_0 + \varepsilon$ and $\varepsilon \geq 1/N$, the corrected margin ε_* is $\Theta(\varepsilon)$ as N grows; near rounding one must retain its explicit formula. No intrinsic lower bound on lists or MCA exceptions follows from this theorem.

9 The optimal high-rate shape and a strict curve improvement

The high-rate problem also has an explicit global solution, but its endpoints need not be constant.

Theorem 9.1 (Optimal high-rate shape). *Suppose $1/2 < \rho < 1$ and $\rho < a < \sqrt{\rho}$. Put*

$$c = a/\rho - 1, \quad d = (a - \rho)/(1 - \rho).$$

The global optimum is

$$\max_{0 \leq B \leq d} \left\{ \left(\frac{a^2}{2\rho} - \frac{1}{2} \right) B + \frac{1-a}{2} B^2 - \frac{2-\rho}{6} B^3 + \frac{\rho}{6(2\rho-1)} (B-c)_+^3 \right\}. \quad (18)$$

An optimal support has decreasing endpoints

$$R_*(s) = \min \left(\frac{a}{\rho}, \frac{2a-1-s}{2\rho-1} \right), \quad 0 \leq s \leq B.$$

Proof. Set $h_0 = (1-a)/(1-\rho) = 1-d$. Since $a < \sqrt{\rho} < (1+\rho)/2$, we have $h_0 > 1/2$ and $0 < c < d < 1/2$. The density is bounded above by the minimum of

$$a-1+(1-\rho)h, \quad a-\frac{1}{2}-\frac{s}{2}-(\rho-1/2)h.$$

Their intersection at $h = 1-s$ has value $a-\rho-(1-\rho)s$. Thus columns $s \geq d$ are nonpositive and may be deleted. Columns ending below h_0 form another nonpositive suffix, since the first expression is negative throughout them.

Every remaining column ends at or above h_0 . The function R_* decreases, lies between h_0 and L , and exceeds s . All three affine expressions whose minimum is w_s are nonnegative on $[h_0, R_*(s)]$. Above $R_*(s)$, if that interval exists below L , the second expression is negative. Moving every remaining endpoint to $R_*(s)$ therefore increases surplus and preserves feasibility. Only the prefix width remains to be chosen.

For $s \leq c$, the column surplus is

$$f(s) = \frac{a^2}{2\rho} - \frac{1}{2} + (1-a)s - \frac{2-\rho}{2}s^2.$$

For $c \leq s \leq d$, integration to the stationary endpoint gives

$$M(s) = \frac{1-\rho}{2} \left[\frac{(d-s)^2}{2\rho-1} - (1-d-s)^2 \right] = f(s) + \frac{\rho}{2(2\rho-1)} (s-c)^2. \quad (19)$$

Integrating these two expressions proves (18). $\square$

Corollary 9.2 (Strict improvement throughout a rate interval). *For every $8-3\sqrt{6} < \rho < 1$, the continuum dimension comparison has positive surplus at some agreement strictly below*

$$a_{\text{DKT}}(\rho) = \frac{3\rho + 2\sqrt{\rho(5-\rho)(2-\rho)}}{8-\rho}.$$

At every such rate, a finite support and multiplicity achieve a strict improvement in characteristic zero and sufficiently large prime characteristic. No uniform bound on that multiplicity or prime is claimed here.

Proof. Set $a = a_{\text{DKT}}(\rho)$ and $B_0 = 3(1 - a)/(2(2 - \rho))$. This a is the positive root of

$$F_\rho(a) = (8 - \rho)a^2 - 6\rho a + \rho(4\rho - 5).$$

Since $F_\rho(\rho) = -\rho(1 - \rho)(5 - \rho) < 0$ and $F_\rho(\sqrt{\rho}) = 3\rho(1 - \sqrt{\rho})^2 > 0$, we have $\rho < a < \sqrt{\rho}$. Also $0 < B_0 < 1/2$ for $\rho > 1/2$. Let $c = a/\rho - 1$. The saturated cap polynomial is

$$P(B) = \left(\frac{a^2}{2\rho} - \frac{1}{2}\right)B + \frac{1-a}{2}B^2 - \frac{2-\rho}{6}B^3.$$

The quadratic equation gives $P(B_0) = 0$.

The actual untrimmed cap $0 \leq s \leq B_0$, $s \leq h \leq a/\rho$ has surplus

$$P(B_0) + \frac{(B_0 - c)_+^3}{12}.$$

Indeed, for $s > c$ the endpoint a/ρ truncates the last triangular rank region, removing rank $(s - c)^2/4$ per column. Integration gives the displayed correction.

It remains to check $B_0 > c$. This is equivalent to $a < T = \rho(7 - 2\rho)/(4 + \rho)$. The polynomial F_ρ is strictly increasing for arguments at least ρ , and $T > \rho$. Direct substitution yields

$$F_\rho(T) = -\frac{4\rho(\rho - 2)(\rho - 1)(\rho^2 - 16\rho + 10)}{(\rho + 4)^2} > 0$$

for $8 - 3\sqrt{6} < \rho < 1$. Therefore $T > a$ and the cap has strictly positive surplus. By continuity, a cap remains positive at some smaller agreement, still above ρ . Fine integer-grid approximations of this fixed positive-area cap give finite positive-surplus supports: normalized rank and benefit converge to their continuum integrals. The exact rank formula applies in characteristic zero or in primes exceeding the chosen finite jet degree. $\square$

The interval improvement and the shape formula above are recovered from the September 13 draft and audited here; they are not claimed as new discoveries of this continuation. The following subsection supplies the explicit cubic formula and its admissible root branch. The following finite certificates give independently checkable instances with explicit multiplicity and prime-characteristic degree guards.

9.1 The explicit continuum threshold

The high-rate shape theorem determines the threshold, including its correct algebraic branch. Write $\rho_* = 8 - 3\sqrt{6}$.

Theorem 9.3 (Sharp continuum threshold above half rate). *For $1/2 < \rho \leq \rho_*$, the threshold is $a_{\text{DKT}}(\rho)$. For $\rho_* < \rho < 1$, put*

$$p = \sqrt{2\rho - 1}, \quad \delta = 1 - p, \quad \gamma = \sqrt{\delta} \frac{(1 + p)(2 - p)}{1 + p^2}.$$

Let y be the largest real root of $y^3 - 3y + \gamma = 0$. Then the threshold is

$$B_* = \frac{p\sqrt{\delta}}{(1 + p^2)y - \delta^{3/2}}, \quad a_* = \frac{1 + p}{2} + \frac{\delta^2}{2}B_*. \quad (20)$$

It satisfies $a_ < a_{\text{DKT}}(\rho)$. At the threshold a nonempty support has zero surplus; below it every positive-area support has negative surplus, and above it positive surplus is possible. These are continuum assertions, not finite-multiplicity converses at high rates.*

Proof. First set $a = a_{\text{DKT}}(\rho)$ and $B_0 = 3(1 - a)/(2(2 - \rho))$. The saturated cap polynomial and its derivative factor as

$$P(B) = -\frac{2-\rho}{6}B(B-B_0)^2, \quad f(s) = -\frac{2-\rho}{2}(s-B_0/3)(s-B_0).$$

The substitution calculation in the preceding corollary also proves $c \geq B_0$ when $1/2 < \rho \leq \rho_*$. In that case $P(B) \leq 0$ for $B \leq c$. On the remaining interval $[c, d]$, the optimized column surplus M of (19) is convex, with $M(c) = f(c) \leq 0$ and $M(d) < 0$. It is therefore nonpositive throughout that interval. No width gives positive surplus, while $B = B_0$ gives zero. Increasing agreement strictly improves the benefit of this nonempty support. A zero-surplus positive-area support at any smaller agreement would similarly give positive surplus before this threshold, a contradiction. This proves the first case.

Now suppose $\rho > \rho_*$. At $a_{\text{low}} = (1 + p)/2$ one has $d = p/(1 + p)$. For $0 < s \leq d$,

$$d - s < p(1 - d - s),$$

so $M(s) < 0$; the full-column surplus satisfies $f(s) \leq M(s)$. Thus every nonempty optimized prefix has negative surplus there. There is positive surplus below a_{DKT} by the preceding corollary. The derivative of total surplus at $B = 0$ is $a^2/(2\rho) - 1/2$, uniformly negative on $[a_{\text{low}}, a_{\text{DKT}}]$. Widths approaching zero therefore cannot supply the first positive surplus. Continuity and compactness give a first threshold strictly between these two agreements, with a nonempty zero-surplus maximizing width B .

At this threshold, $B > c$: below a_{DKT} the saturated polynomial $P(B)$ is strictly negative for every $B > 0$. Also $B < d$, since $M(d) < 0$ would make a slightly smaller width have positive surplus if $B = d$ had zero surplus. Thus stationarity gives $M(B) = 0$. Both $d - B$ and $1 - d - B$ are positive, so this equation has the definite sign choice

$$d - B = p(1 - d - B), \quad a = \frac{1+p}{2} + \frac{(1-p)^2}{2}B.$$

Put $x = c/B$. Then

$$B = \frac{p\delta}{(1+p^2)x - \delta^2}, \quad Q(x) = x^3 - 3\delta x + \delta^2 \frac{(1+p)(2-p)}{1+p^2}.$$

Substitution into the total surplus of (18) gives exactly

$$-\frac{p\delta^3(1+p^2)}{12((1+p^2)x - \delta^2)^3} Q(x).$$

Hence $Q(x) = 0$.

For the branch choice, $B < d < 1/2$ implies $x > x_{1/2} = (1 - p^2)/(1 + p^2)$, and

$$Q(0) > 0, \quad Q(x_{1/2}) = -\frac{p^2(1-p)^2(1+p)^2(p^2+3)}{(1+p^2)^3} < 0.$$

The cubic has two positive roots, exactly one greater than $x_{1/2}$; its third root is negative. Thus the threshold selects the largest root. Setting $x = \sqrt{\delta} y$ proves (20). Existence and admissibility of this root follow from the interior optimizer just established, rather than from an unsupported choice of a numerical cubic branch.

Finally, the threshold support gains strictly when agreement increases. A positive-area zero-surplus support below threshold would also gain strictly at a slightly higher agreement still below threshold, which is impossible. This proves the stated interpretation on both sides. $\square$

The formula and proof are recovered from the September 13 draft, with an expanded branch and existence audit here. Exact symbolic checks verify the three algebraic identities used above. Rational isolation at six rational values of p produces admissible root intervals and agreement intervals of width less than 10^{-28} , with opposite surplus signs at their endpoints. These checks supplement the proof; they are not a substitute for the quantified threshold statement.

10 Exact finite certificates for the high-rate improvement

The sufficient first-order curve in the recovered Dao–Kominers–Thaler working draft [1], on the high-rate branch used here, is

$$a_{\text{DKT}}(\rho) = \frac{3\rho + 2\sqrt{\rho(5-\rho)(2-\rho)}}{8-\rho}. \quad (21)$$

The following two exact supports improve this sufficient curve without requiring a continuum optimality claim at those rates. Choose m, ρ, a, B and set

$$q_b = \min\left(\left\lceil \frac{ma}{\rho} \right\rceil - 1, \left\lfloor \frac{2ma - m - b}{2\rho - 1} \right\rfloor\right), \quad 0 \leq b \leq \lfloor mB \rfloor, \quad H_b = q_b - b + 1.$$

These endpoints decrease and all selected monomials have positive benefit. Their exact rank can be summed columnwise in linear time. Put

$$C(t) = \left\lfloor \frac{(\max(0, t) + 1)^2}{4} \right\rfloor, \quad D(t, b) = \left\lfloor \frac{\max(0, t - 2b)^2}{4} \right\rfloor.$$

Then column b has rank

$$C(m) - C(\max(0, m - H_b)) + D(m, b) - D(\max(0, m - H_b), b).$$

This is the sum of (5), using

$$d_m(u, b) = \left\lceil \frac{(m - u)_+}{2} \right\rceil + \left(\left\lfloor \frac{m - u}{2} \right\rfloor - b \right)_+.$$

	First certificate	Second certificate
ρ	9/10	3/4
a	47389/50000	43049/50000
m	4096	65536
B	49/500	173/1000
Columns	402	11338
Monomials	1632467	786044039
Exact rank	3061838715	20619038985535
Benefit	9568416878978/3125	128869578783487787/6250
Surplus	170894603/3125	585123894037/6250

Both agreements are strictly below (21). The displayed counts are exact rational arithmetic; the large supports are specified by their endpoint formulas rather than enumerated.

To relate a positive surplus to finite block length, take a message-degree cap $D \leq \rho N - 1$ and agreement target $A \geq aN$. The permitted X coefficients of $Y_0^u Y_1^b$ under weighted degree $mA - 1$ number

$$mA - Du - (D - 1)b \geq N(ma - \rho(u + b)).$$

Here is explicit challenge-degree bookkeeping. Let $q_{\max}$ be the largest source jet degree, let r_m be its saturated local rank, and let G be the total number of permitted X coefficients. Recentring Y_0 by a received value affine in the challenge gives local equations of challenge degree at most $q_{\max}$. A fixed basis of the centered local row space supplies r_m such equations per coordinate. Giving every global coefficient challenge degree at most ℓ therefore gives $(\ell + 1)G$ unknowns and at most $Nr_m(\ell + q_{\max} + 1)$ scalar constraints. Since $G \geq NB_{m,\rho,a}$, it suffices to take

$$\ell + 1 > \frac{q_{\max}r_m}{B_{m,\rho,a} - r_m}.$$

This is a fixed finite budget for either certificate. The resulting nonzero interpolation polynomial has weighted degree at most $mA - 1$; substitution of a candidate with A agreements gives at least mA zeros counted with multiplicity and hence an identity. The subsequent candidate and incidence arguments are the usual first-order ones in sufficiently large characteristic; the result improves the agreement curve, not the exponent of the exceptional-count bound.

11 Verification and scope

The exact verifiers check 320 local matrices, 3750 rearrangement and marginal comparisons, and 3128 one-sided column inequalities by rational polygon integration. Independent diagonal sums reproduce the finite certificates after 21295 direct checks of the summation formula. Further checks cover 1920 finite-cutoff ranks, 32752 half-rank instances, 3743 arbitrary-support rank estimates, the narrow-cap obstruction, and three positive-surplus parameter certificates. Nonmonomial degeneration passes 504 rank comparisons, including 196 strict drops. The finite-length cost transfer passes 6750 reductions, including 1054 positive-surplus ratio comparisons and a near-critical certificate at length one million. The high-rate shape passes 220 exact polygon integrations and 4851 endpoint comparisons. These scripts and reports are in `research/first_order_support_audit/`.

The current graded-cost adaptation additionally passes 79002 algebraic reconstruction-degree checks and 16681 low-degree rank and negative-area checks for multiplicities 16 through 512, recorded in `research/current_graded_method_costs/`. The computations corroborate formulas; the proofs establish the quantified statements.

The sufficient challenge degrees obtained by the displayed conservative count are 241481265 and 16569483999 for the two high-rate certificates. These are not claimed to be necessary. Together with the large supports, they make clear that a strict mathematical curve improvement is not a practical parameter improvement for a deployed proof system.

The fixed-jet-space nonmonomial theorem and the exact finite-length monomial theorem have different scopes. Neither proves that a global interpolation kernel is absent when the dimension count fails: dependencies between coordinates may reduce the global constraint rank. Neither proves a lower bound on actual list size, nor necessity of the quadratic exceptional-count bound for first-order MCA.

References

- [1] Q. Dao, S. D. Kominers, and J. Thaler. *Quantitative Reed–Solomon List Decoding and Mutual Correlated Agreement: From Johnson to Capacity*. September 2026 working draft. The sufficient curve used here was checked in the recovered 105-page version dated September 10; this does not assert it is the latest version.

- [2] Q. Dao, S. D. Kominers, and J. Thaler. *Reed–Solomon Codes Beyond Johnson: Efficient Decoding and Smaller Cryptographic Proofs*. Cryptology ePrint Archive, Report 2026/2056. 145-page version retrieved September 17, 2026. <https://eprint.iacr.org/2026/2056>.